

Fourier Series

$f(x)$ is a periodic function in $(c, c+2l)$ then it can be expanded as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{l}x\right)$$

$$a_0 = \frac{1}{l} \int_c^{c+2l} f(x) dx$$

$$a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos\left(\frac{n\pi}{l}x\right) dx$$

$$b_n = \frac{1}{l} \int_c^{c+2l} f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

} Fourier coefficients

Type-I : Interval $(0, 2\pi)$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos(nx) + \sum b_n \sin(nx)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx$$

Q1. Find F.S for $f(x) = x^2$ in the interval $(0, 2\pi)$

Hence deduce that,

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$$

$$\text{Sol}^n f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} x^2 dx = \frac{1}{\pi} \left(\frac{x^3}{3} \right)_0^{2\pi}$$

$$= \frac{1}{\pi} \left[\frac{8\pi^3}{3} - 0 \right] = \frac{8\pi^2}{3}$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \cos(nx) dx$$

$$\int u \cdot v = u \cdot v - u'v_2 + u''v_3 + \dots$$

$$a_n = \frac{1}{\pi} \left\{ x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right\}_0^{2\pi}$$

$$= \frac{2}{\pi n^2} \left\{ 2\pi \cos(2n\pi) - 0 \right\}$$

$$\cos(nx) = (-1)^n$$

$$= \frac{4\pi}{\pi n^2} = \frac{4}{n^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_0^{2\pi} x^2 \sin nx dx$$

$$= \frac{1}{\pi} \left\{ x^2 \left(\frac{-\cos nx}{n} \right) - 2x \left(\frac{-\sin nx}{n^2} \right) + 2 \left(\frac{\cos nx}{n^3} \right) \right\}_0^{2\pi}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{-4\pi^2 \cos(2n\pi)}{n} - 0 \right) + \frac{2}{n^3} (\cos(2n\pi) - \cos 0) \right\}$$

$$= \frac{1}{\pi} \cdot \frac{-4\pi^2}{n}$$

$$b_n = -\frac{4\pi}{n}$$

$$\therefore f(x) = \frac{4\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos(nx) + \sum_{n=1}^{\infty} \left(\frac{-4\pi}{n} \right) \sin(nx)$$

$$= \frac{4\pi^2}{3} + \left[\frac{4}{1^2} \cos x + \frac{4}{2^2} \cos 2x + \frac{4}{3^2} \cos 3x - \dots \right] + \left[\frac{-4\pi}{1} \sin x - \frac{4\pi}{2} \sin 2x - \frac{4\pi}{3} \sin 3x - \dots \right]$$

put $x = \pi$,

$$\pi^2 = \frac{4\pi^2}{3} + \left[\frac{4}{1^2} (-1) + \frac{4}{2^2} (1) + \frac{4}{3^2} (-1) + \frac{4}{4^2} (1) - \dots \right]$$

$$\pi^2 - \frac{4\pi^2}{3} = 4 \left[\frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots \right]$$

$$\frac{-\pi^2}{3 \times 4} = \frac{-1}{1^2} + \frac{1}{2^2} - \frac{1}{3^2} + \frac{1}{4^2} - \dots$$

$$\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$$

Q2. Find F.S for $f(x) = e^{-x}$ in $(0, 2\pi)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{-1} \right]_0^{2\pi} = \frac{1}{\pi} \left[\frac{e^{-2\pi} - 1}{-1} \right] = \frac{1}{\pi} (1 - e^{-2\pi})$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \cos(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{(-1)^2 + n^2} (-\cos(nx) + n \sin(nx)) \right]_0^{2\pi}$$

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$= \frac{1}{\pi(1+n^2)} \{ e^{-2\pi} \cos(2n\pi) - (-e^0 \cos 0) \}$$

$$= \frac{1}{\pi(1+n^2)} \{ e^{-2\pi} + 1 \}$$

$$a_n = \frac{1 - e^{-2\pi}}{\pi(1+n^2)}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \int_0^{2\pi} e^{-x} \sin(nx) dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-x}}{(-1)^2 + n^2} (-\sin(nx) - n \cos(nx)) \right]_0^{2\pi}$$

$$= \frac{1}{\pi(1+n^2)} \{ -n e^{-2\pi} \cos(2n\pi) - (-n e^0 \cos 0) \}$$

$$= \frac{1}{\pi(1+n^2)} \{ -n e^{-2\pi} + n \}$$

$$b_n = \frac{n(1 - e^{-2\pi})}{\pi(1+n^2)}$$

The F.S is:

$$e^{-x} = \frac{1 - e^{-2\pi}}{2\pi} + \sum_{n=1}^{\infty} \frac{(1 - e^{-2\pi}) \cos(nx)}{\pi(1+n^2)} + \sum_{n=1}^{\infty} \frac{n(1 - e^{-2\pi}) \sin(nx)}{\pi(1+n^2)}$$

$$\text{Q3 } f(x) = \begin{cases} x & , 0 < x \leq \pi \\ 2\pi - x & , \pi \leq x \leq 2\pi \end{cases}$$

$$\text{Hence deduce that } \frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

$$\text{Sol}^n f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x dx + \int_{\pi}^{2\pi} (2\pi - x) dx \right\}$$

$$a_0 = \frac{1}{\pi} \left\{ \left(\frac{x^2}{2} \right)_0^{\pi} + \left(2\pi x - \frac{x^2}{2} \right)_{\pi}^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{\pi^2}{2} - 0 \right) + \left[\left(4\pi^2 - \frac{4\pi^2}{2} \right) - \left(2\pi^2 - \frac{\pi^2}{2} \right) \right] \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{\pi^2}{2} + 2\pi^2 - 2\pi^2 + \frac{\pi^2}{2} \right\}$$

$$a_0 = \pi$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos(nx) dx$$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x \cos(nx) dx + \int_{\pi}^{2\pi} (2\pi - x) \cos(nx) dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[x \left(\frac{\sin nx}{n} \right) - (-1) \left(\frac{-\cos nx}{n^2} \right) \right]_0^{\pi} \right. \\ \left. + \left[(2\pi - x) \left(\frac{\sin nx}{n} \right) - (-1) \left(\frac{-\cos nx}{n^2} \right) \right]_0^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \frac{1}{n^2} (\cos n\pi - \cos 0) - \frac{1}{n^2} (\cos n 2\pi - \cos n\pi) \right\}$$

$$= \frac{1}{\pi n^2} \{ (-1)^n - 1 - 1 + (-1)^n \}$$

$$a_n = \frac{2[(-1)^n - 1]}{\pi n^2} = \begin{cases} 0, & n \text{ is even} \\ -\frac{4}{\pi n^2}, & n \text{ is odd} \end{cases}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin(nx) dx$$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x \sin(nx) dx + \int_{\pi}^{2\pi} (2\pi - x) \sin(nx) dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left[x \left(\frac{-\cos nx}{n} \right) - (-1) \left(\frac{-\sin nx}{n^2} \right) \right]_0^{\pi} \right. \\ \left. + \left[(2\pi - x) \left(\frac{-\cos nx}{n} \right) - (-1) \left(\frac{-\sin nx}{n^2} \right) \right]_{\pi}^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{-\pi \cos n\pi}{n} - 0 \right) + \left(0 - (2\pi) \left(\frac{-\cos n\pi}{n} \right) \right) \right\}$$

$$b_n = 0$$

F.S for $f(x)$ is,

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2[(-1)^n - 1]}{\pi n^2} \cos(nx)$$

$$\frac{4\pi^2}{2}$$

Parseval's Identity: The trig series has double power, we use this

$$\frac{1}{\pi} \int_0^{2\pi} [f(x)]^2 dx = \frac{a_0^2}{2} + \sum (a_n^2 + b_n^2)$$

$$\text{LHS} = \frac{1}{\pi} \int_0^{2\pi} [f(x)]^2 dx$$

$$= \frac{1}{\pi} \left\{ \int_0^{\pi} x^2 dx + \int_{\pi}^{2\pi} (2\pi - x)^2 dx \right\}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{x^3}{3} \right)_0^{\pi} + \left(\frac{(2\pi - x)^3}{-3} \right)_{\pi}^{2\pi} \right\}$$

$$= \frac{1}{\pi} \left\{ \left(\frac{\pi^3}{3} - 0 \right) + \left(0 - \frac{\pi^3}{-3} \right) \right\}$$

$$\frac{2\pi^2}{3} = \frac{\pi^2}{2} + \sum \left[\frac{2[(-1)^n - 1]^2}{\pi n^2} \right]$$

$$\frac{2\pi^2}{3} - \frac{\pi^2}{2} = \sum \frac{4[(-1)^n - 1]^2}{\pi^2 n^4}$$

$$\frac{\pi^2}{6} = \frac{16}{\pi^2 \cdot 1^4} + \frac{16}{\pi^2 \cdot 3^4} + \frac{16}{\pi^2 \cdot 5^4} + \dots$$

$$= \frac{16}{\pi^2} \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right]$$

$$\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

Type-2: F.S in $(0, 2L)$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{L}x\right) + \sum b_n \sin\left(\frac{n\pi}{L}x\right)$$

$$a_0 = \frac{1}{2L} \int_0^{2L} f(x) dx$$

$$a_n = \frac{1}{L} \int_0^{2L} f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

$$b_n = \frac{1}{L} \int_0^{2L} f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Q1. Find F.S for $f(x) = 4 - x^2$ in $(0, 2)$

Solⁿ $(0, 2) \leftrightarrow (0, 2L)$

$$\boxed{L=1}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum a_n \cos(n\pi x) + \sum b_n \sin(n\pi x)$$

$$a_0 = \frac{1}{L} \int_0^{2L} f(x) dx = \int_0^2 (4 - x^2) dx$$

$$= \left(4x - \frac{x^3}{3}\right)_0^2 = \left(8 - \frac{8}{3}\right) = \frac{16}{3}$$

$$a_n = \frac{1}{L} \int_0^{2L} f(x) \cos\left(\frac{n\pi}{L}x\right) dx = \int_0^2 (4 - x^2) \cos(n\pi x) dx$$

$$= \left\{ (4 - x^2) \left(\frac{\sin(n\pi x)}{n\pi} \right) - (-2x) \left(\frac{-\cos(n\pi x)}{n^2 \pi^2} \right) + (-2) \left(\frac{-\sin(n\pi x)}{n^3 \pi^3} \right) \right\}_0^2$$

$$= -\frac{2}{n^2 \pi^2} \left[2 \cos(2n\pi) - 0 \right]$$

$$\boxed{a_n = -\frac{4}{n^2 \pi^2}}$$

$$b_n = \frac{1}{L} \int_0^{2L} f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{1}{L} \int_0^2 (4 - x^2) \sin(n\pi x) dx$$

$$= \frac{1}{L} \int_0^2 (4 - x^2) \sin(n\pi x) dx$$

$$= \left\{ (4 - x^2) \left(\frac{-\cos(n\pi x)}{n\pi} \right) - (-2x) \left(\frac{-\sin(n\pi x)}{n^2 \pi^2} \right) + (-2) \left(\frac{\cos(n\pi x)}{n^3 \pi^3} \right) \right\}_0^2$$

$$= \left\{ \left[0 - 4 \left(\frac{-\cos 0}{n\pi} \right) \right] - \frac{2}{n^3 \pi^3} [\cos 2n\pi - \cos 0] \right\}$$

$$\boxed{b_n = \frac{4}{n\pi}}$$

$$f(x) = \frac{8}{3} + \sum_{n=1}^{\infty} -\frac{4}{n^2 \pi^2} \cos(n\pi x) + \sum_{n=1}^{\infty} \frac{4}{n\pi} \sin(n\pi x)$$

Q2. Find F.S for

$$f(x) = \begin{cases} 1 & 0 < x \leq 1 \\ x & 1 < x \leq 2 \end{cases}$$

H.W

$$l = 1$$

$$\therefore a_0 = \frac{1}{2l} \int_0^{2l} f(x) dx$$

$$= \frac{1}{2} \left[\int_0^1 1 dx + \int_1^2 x dx \right]$$

$$= \frac{1}{2} \left[(x)_0^1 + \left(\frac{x^2}{2} \right)_1^2 \right]$$

$$= \frac{1}{2} \left[1 + \left(\frac{4}{2} - \frac{1}{2} \right) \right]$$

$$= \frac{1}{2} \left[1 + \frac{3}{2} \right]$$

$$= \frac{1}{2} \times \frac{5}{2} = \frac{5}{4}$$

$$a_n = \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{1} \int_0^1 1 \cdot \cos n\pi x dx + \int_1^2 x \cdot \cos n\pi x dx$$

$$= \left[\frac{\sin(n\pi x)}{n\pi} \right]_0^1 + \left[x \cdot \frac{\sin(n\pi x)}{n\pi} - (1) \left(\frac{-\cos(n\pi x)}{n^2 \pi^2} \right) \right]_1^2$$

$$= \left[\frac{\cos(n\pi x)}{n^2 \pi^2} \right]_1^2$$

$$= \frac{\cos 2n\pi}{n^2 \pi^2} - \frac{\cos n\pi}{n^2 \pi^2} = \frac{1}{n^2 \pi^2} (1 - (-1)^n)$$

$$= \frac{1 - (-1)^n}{n^2 \pi^2} \begin{cases} 0, & n \text{ is even} \\ \frac{2}{n^2 \pi^2}, & n \text{ is odd} \end{cases}$$

$$b_n = \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \left[\frac{-\cos(n\pi x)}{n\pi} \right]_0^1 + \left[x \cdot \left(\frac{-\cos(n\pi x)}{n\pi} \right) - 1 \cdot \left(\frac{-\sin(n\pi x)}{n^2 \pi^2} \right) \right]_1^2$$

$$= \frac{1}{n\pi} [-\cos(n\pi) + \cos 0] + \left[2 \cdot \left(\frac{-\cos 2n\pi}{n\pi} \right) + \frac{\cos n\pi}{n\pi} \right]$$

$$= \frac{1}{n\pi} [-(-1)^n + 1] + \left[\frac{-2}{n\pi} + \frac{(-1)^n}{n\pi} \right]$$

$$= \frac{[1] - (-1)^n}{n\pi} + \frac{1}{n\pi} [-2 + (-1)^n]$$

$$= \frac{1}{n\pi} [-(-1)^n + 1 - 2 + (-1)^n]$$

$$= \frac{1}{n\pi} [(-1)^n - (-1)^n - 1]$$

$$= -\frac{1}{n\pi}$$

$$f(x) = \frac{5}{4} + \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^2 \pi^2} \cos(n\pi x) - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin(n\pi x)$$

$$\text{p3 } f(x) = \frac{1-x}{2}, \quad 0 \leq x < 1$$

$$\text{Sol}^n (0, 1) \leftrightarrow (0, 2l)$$

$$\therefore l = 1/2$$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi x}{l}\right) + \sum b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{a_0}{2} + \sum a_n \cos(2n\pi x) + \sum b_n \sin(2n\pi x)$$

$$a_0 = \frac{1}{l_0} \int_0^{2l} f(x) dx = 2 \int_0^1 \left(\frac{1}{2} - x\right) dx$$

$$= 2 \left[\frac{x}{2} - \frac{x^2}{2} \right]_0^1$$

$$= 1 [0 - 0]$$

$$a_0 = 0$$

$$a_n = \frac{1}{l_0} \int_0^{2l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx = 2 \int_0^1 \left(\frac{1}{2} - x\right) \cos(2n\pi x) dx$$

$$= 2 \left\{ \left(\frac{1}{2} - x\right) \frac{\sin(2n\pi x)}{2n\pi} - (-1) \left(\frac{-\cos(2n\pi x)}{(2n\pi)^2}\right) \right\}_0^1$$

$$= -\frac{2}{24n^2\pi^2} [\cos(2n\pi) - \cos 0]$$

$$a_n = 0$$

$$b_n = \frac{1}{l_0} \int_0^{2l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx = 2 \int_0^1 \left(\frac{1}{2} - x\right) \sin(2n\pi x) dx$$

$$= 2 \left\{ \left(\frac{1}{2} - x\right) \frac{-\cos(2n\pi x)}{2n\pi} - (-1) \left(\frac{-\sin(2n\pi x)}{(2n\pi)^2}\right) \right\}_0^1$$

$$= 2 \left\{ \left(\frac{1}{2} - 1\right) \frac{-\cos(2n\pi)}{2n\pi} - \left(\frac{1}{2}\right) \frac{-\cos 0}{2n\pi} \right\}$$

$$= 2 \left\{ \frac{+1 \times 1}{2 \cdot 2n\pi} + \frac{1 \times 1}{2 \cdot 2n\pi} \right\}$$

$$= 2 \left\{ \frac{1}{4n\pi} + \frac{1}{4n\pi} \right\}$$

$$b_n = \frac{1}{2n\pi}$$

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin(2n\pi x)$$

Type-3/Type-4 $(-l, l)/(-\pi, \pi)$

Even function:

$$g.f. f(-x) = f(x)$$

$$\text{eg: } x^2, x^4, x^6, \dots$$

$$\cos x, |x|$$

$$E \pm E = E$$

$$O \pm O = O$$

$$E \pm O = N.E, N.O$$

$$E \times E = E$$

$$O \times O = E$$

$$O \times E = O$$

Odd function:

$$f(-x) = -f(x)$$

$$\text{eg: } x, x^3, x^5, \dots$$

$$\sin x$$

$$\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is even} \\ 0, & \text{if } f(x) \text{ is odd} \end{cases}$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \begin{cases} \frac{2}{l} \int_0^l f(x) dx & \text{if } f(x) \text{ is even} \\ 0 & \text{if } f(x) \text{ is odd} \end{cases}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx = \begin{cases} \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx, & \text{if } f(x) \text{ is even} \\ 0, & \text{if } f(x) \text{ is odd} \end{cases}$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx = \begin{cases} 0, & \text{if } f(x) \text{ is even} \\ \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx, & \text{if } f(x) \text{ is odd} \end{cases}$$

① If $f(x)$ is even in $(-l, l)$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l}x\right) dx$$

$$b_n = 0$$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right)$$

② If $f(x)$ is odd in $(-l, l)$

$$a_0 = 0, a_n = 0$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$f(x) = \sum b_n \sin\left(\frac{n\pi}{l}x\right)$$

Q1. $f(x) = x^3$ in $(-\pi, \pi)$

Solⁿ: $f(-x) = f(-x^3) = -x^3 = -f(x)$
 $\therefore f(x)$ is an odd function
 $\therefore a_0 = 0, a_n = 0$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$= \frac{2}{\pi} \int_0^\pi x^3 \sin(nx) dx$$

$$= \frac{2}{\pi} \left\{ x^3 \left(\frac{-\cos(nx)}{n} \right) - (3x^2) \left(\frac{-\sin(nx)}{n^2} \right) + (6x) \left(\frac{\cos(nx)}{n^3} \right) - (6) \left(\frac{\sin(nx)}{n^4} \right) \right\}_0^\pi$$

$$= \frac{2}{\pi} \left\{ -\frac{\pi^3 \cos n\pi}{n} + \frac{6\pi \cos(n\pi)}{n^3} \right\}$$

$$= \frac{2}{\pi} \pi (-1)^n \left\{ -\frac{\pi^2}{n} + \frac{6}{n^3} \right\}$$

$$b_n = 2(-1)^n \left\{ \frac{6}{n^3} - \frac{\pi^2}{n} \right\}$$

$$f(x) = \sum b_n \sin\left(\frac{n\pi}{l}x\right)$$

$$x^3 = \sum_{n=1}^{\infty} 2(-1)^n \left[\frac{6}{n^3} - \frac{\pi^2}{n} \right] \sin(nx)$$

Ex-2:

Q. $f(x) = 1 - x^2$ in $(-1, 1)$ Find F.S

Solⁿ: $f(-x) = 1 - (-x)^2 = 1 - x^2 = f(x)$
 $\therefore f(x)$ is an even function

$$b_n = 0$$

$$(-1, 1) \leftrightarrow (-l, l) \Rightarrow l = 1$$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right)$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x)$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = 2 \int_0^1 (1 - x^2) dx$$

$$= 2 \left[x - \frac{x^3}{3} \right]_0^1$$

$$a_0 = 2 \left[1 - \frac{1}{3} \right]$$

$$a_0 = \frac{4}{3}$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= 2 \int_0^1 (1-x^2) \cos(n\pi x) dx$$

$$= 2 \left[(1-x^2) \frac{\sin(n\pi x)}{n\pi} - (-2x) \left(-\frac{\cos(n\pi x)}{n^2\pi^2} \right) + 2 \left(-\frac{\sin(n\pi x)}{n^3\pi^3} \right) \right]_0^1$$

$$= 2 \left[-2x \left(\frac{\cos(n\pi x)}{n^2\pi^2} \right) \right]_0^1$$

$\Rightarrow \frac{4}{n^2\pi^2} [(-1)^n - 1]$ if n is odd
0, if n is even

$$= \frac{-4}{n^2\pi^2} (-1)^n$$

$\therefore$ F.S for $f(x) = 1-x^2$ in $(-1, 1)$ is

$$1-x^2 = \frac{2}{3} + \sum \frac{-4(-1)^n}{n^2\pi^2} \cos(n\pi x)$$

Q. ~~Find~~ $f(x) = x|x|$ in $(-1, 1)$

$x|x| = -x^2 \quad -1 < x < 0$
 $x^2 \quad 0 < x < 1$

$f(-x) = (-x)|-x| = -x|x| = -f(x)$

$\therefore f(x)$ is an odd function
 $a_0 = 0, a_n = 0$

$$f(x) = \sum b_n \sin\left(\frac{n\pi}{l}x\right)$$

$$= \sum b_n \sin(n\pi x)$$

$$= \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

$$= \frac{2}{l} \int_0^1 x^2 \sin(n\pi x) dx$$

Q Find F.S for $f(x) = e^x$ in $(-1, 1)$

Solⁿ $(-1, 1) \leftrightarrow (-1, 1)$
 $\Rightarrow l = 1$

$$f(x) = e^x$$

$$f(-x) = e^{-x}$$

$$\therefore f(-x) \neq f(x) \text{ and } f(-x) \neq -f(x)$$

$\therefore f(x)$ is a N.E.N.O fn.

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right) + \sum b_n \sin\left(\frac{n\pi}{l}x\right)$$

$$= \frac{a_0}{2} + \sum a_n \cos n\pi x + \sum b_n \sin n\pi x$$

$$a_0 = \frac{1}{l} \int_{-1}^1 f(x) dx$$

$$= \frac{1}{1} \int_{-1}^1 e^x dx = \left[e^x \right]_{-1}^1$$

$$= e' - e^{-1}$$

$$= e - \frac{1}{e}$$

$$a_0 = \frac{e^2 - 1}{e}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{e} \int_{-1}^1 e^x \cos(n\pi x) dx$$

$$= \left\{ \frac{e^{ax}}{1^2 + (n\pi)^2} \left[(1) \cos(n\pi x) + n\pi \sin(n\pi x) \right] \right\}_{-1}^1$$

$$= \frac{e}{1+n^2\pi^2} [\cos n\pi] - \frac{e^{-1}}{1+n^2\pi^2} [\cos n\pi]$$

$$= \frac{(-1)^n}{1+n^2\pi^2} [e - e^{-1}]$$

$$= \frac{(-1)^n}{1+n^2\pi^2} \left[e - \frac{1}{e} \right]$$

$$a_n = \frac{(-1)^n}{1+n^2\pi^2} \left[\frac{e^2 - 1}{e} \right]$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \int_{-1}^1 e^x \sin(n\pi x) dx$$

$$= \frac{e^n}{1+n^2\pi^2} \left[(1) \sin n\pi x - n\pi \cos(n\pi x) \right]_{-1}^1$$

$$= \frac{e}{1+n^2\pi^2} [-n\pi \cos n\pi] - \frac{e^{-1}}{1+n^2\pi^2} [-n\pi \cos n\pi]$$

$$= \frac{-e^{-1}}{1+n^2\pi^2} [e - e^{-1}]$$

$$= \frac{-n\pi(-1)^n}{1+n^2\pi^2} \left[\frac{1 - e^2}{e} \right]$$

$\therefore$ F.S for $f(x)$ is

$$e^x = \frac{e^2 - 1}{e} + \sum \frac{(-1)^n}{1+n^2\pi^2} \left(\frac{e^2 - 1}{e} \right) \cos n\pi x + \sum \frac{n\pi(-1)^n}{1+n^2\pi^2} \left(\frac{1 - e^2}{e} \right) \sin(n\pi x)$$

$$f(x) = \begin{cases} 0 & -2 < x < -1 \\ 1+x & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$$

$$\text{Sol}^n: (-1, 1) \leftrightarrow (-2, 2) \\ l = 2$$

$$f(-x) = \begin{cases} 0 & -2 < -x < -1 \\ 1+(-x) & -1 < -x < 0 \\ 1-(-x) & 0 < -x < 1 \\ 0 & 1 < -x < 2 \end{cases}$$

Multiply by -1 in range

$$= \begin{cases} 0 & 2 > x > 1 \\ 1-x & 1 > x > 0 \\ 1+x & 0 > x > -1 \\ 0 & -1 > x > -2 \end{cases} = f(x)$$

$f(x)$ is an even function
 $b_n = 0$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right)$$

$$= \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{2}x\right)$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$= \frac{2}{2} \int_0^2 f(x) dx$$

$$= \int_0^2 (1-x) dx = \left(x - \frac{x^2}{2} \right)_0^2 = 1 - 1 = 0$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l}x\right) dx$$

$$= \frac{2}{2} \int_0^2 f(x) \cos\left(\frac{n\pi}{2}x\right) dx$$

$$= \int_0^2 (1-x) \cos\left(\frac{n\pi}{2}x\right) dx$$

$$= \left\{ (1-x) \left(\frac{\sin\left(\frac{n\pi}{2}x\right)}{\left(\frac{n\pi}{2}\right)} \right) - (-1) \left(\frac{-\cos\left(\frac{n\pi}{2}x\right)}{\left(\frac{n\pi}{2}\right)^2} \right) \right\}_0^2$$

$$a_n = -\frac{4}{n^2 \pi^2} \left[\frac{\cos n\pi}{2} - \cos 0 \right]$$

$$= \frac{4}{n^2 \pi^2} \left[1 - \cos n\pi \right]$$

F.S for $f(x)$ is:

$$f(x) = \frac{1}{4} + \sum \frac{4}{n^2 \pi^2} \left[1 - \cos n\pi \right] \cos\left(\frac{n\pi}{2}x\right)$$

Half range Series

Half range cosine series (HRCs)
 $f(x)$ in $(0, l)$

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right)$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l}x\right) dx$$

Half range sine series (HRSS)
 $f(x)$ in $(0, l)$

$$f(x) = \sum b_n \sin\left(\frac{n\pi}{l}x\right)$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi}{l}x\right) dx$$

Find HRCs for $f(x) = x$ in $(0, 2)$
Hence deduce that

$$\textcircled{1} \frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

$$\textcircled{2} \frac{\pi^4}{90} = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots$$

Solⁿ $(0, 2) \leftrightarrow (0, 1)$
 $l = 2$

HRCS in

$$f(x) = \frac{a_0}{2} + \sum a_n \cos\left(\frac{n\pi}{l}x\right)$$

$$f(x) = a_0 + \sum a_n \cos\left(\frac{n\pi}{2}x\right)$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx = \frac{2}{2} \int_0^2 x dx$$

$$= \left(\frac{x^2}{2}\right)_0^2$$

$$= 2$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi}{l}x\right) dx$$

$$= \frac{2}{2} \int_0^2 x \cos\left(\frac{n\pi}{2}x\right) dx$$

$$= \left\{ x \left(\frac{\sin\left(\frac{n\pi}{2}x\right)}{\left(\frac{n\pi}{2}\right)} \right) - (-1) \left(\frac{-\cos\left(\frac{n\pi}{2}x\right)}{\left(\frac{n\pi}{2}\right)^2} \right) \right\}_0^2$$

$$= \frac{4}{n^2 \pi^2} [\cos(n\pi) - \cos 0] = \frac{4}{n^2 \pi^2} [(-1)^n - 1]$$

HRCS for $f(x) = x$ in $(0, 2)$ is

$$x = 1 + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} [(-1)^n - 1] \cos\left(\frac{n\pi}{2}x\right)$$

By Parseval's identity,

$$\frac{2}{l} \int_0^l (f(x))^2 dx = \frac{a_0^2}{2} + \sum a_n^2$$

$$\frac{2}{2} \int_0^2 x^2 dx = \frac{4}{2} + \sum_{n=1}^{\infty} \frac{16}{n^4 \pi^4} [(-1)^n - 1]^2$$

$$\left(\frac{x^3}{3}\right)_0^2 = 2 + \frac{16}{\pi^4} \left[\frac{4}{1^4} + \frac{4}{3^4} + \frac{4}{5^4} + \dots \right]$$

$$\frac{8}{3} - 2 = \frac{64}{\pi^4} \left[\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots \right]$$

$$\frac{2 \times \pi^4}{3 \times 64} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

$$\frac{\pi^4}{96} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \dots$$

$$\frac{1}{2^4} + \frac{1}{4^4} + \frac{1}{6^4} + \dots$$

$$\frac{1}{2^4 \cdot 1^4}$$

② Let $S = \frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots$

$$S = \left(\frac{1}{1^4} + \frac{1}{3^4} + \dots \right) + \left(\frac{1}{2^4} + \frac{1}{4^4} + \dots \right)$$

$$= \frac{\pi^4}{96} + \frac{1}{2^4} \left[\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \dots \right]$$

$$= \frac{\pi^4}{96} + \frac{1}{16} S$$

$$S - \frac{S}{16} = \frac{\pi^4}{96}$$

$$\Rightarrow \frac{15S}{16} = \frac{\pi^4}{96} \Rightarrow S = \frac{\pi^4}{90}$$